

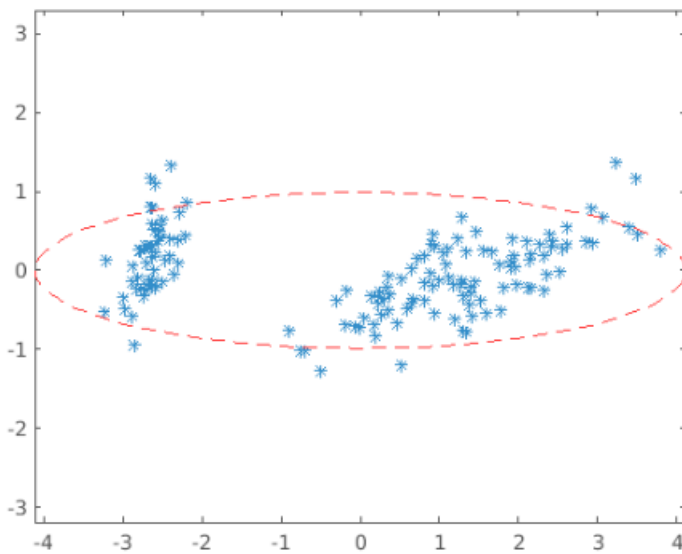
Plotting the ellipsoid fit to the data

Create a script to scatter plot the first 2 dimensions of the PCA linear vector space, and display a contour representing the boundary, in the ellipsoid with 2 standard deviations width.

The steps are:

- Use the 'pca' function to find **D_pca**, the coordinates of the dataset in the PCA vector space, and **eigvals**, the corresponding eigenvalues of the PCA ellipsoid.
- Define the x and y coordinates of the first two dimensions of the data in the PCA vector space, **D_pca_x** and **D_pca_y**. Scatter plot the data using asterisks.
- Define x and y coordinates for the shape of the PCA ellipsoid in just the first two dimensions, which is a standard ellipse, as **ellipse_x** and **ellipse_y**. The long and short axes radii should be 2 standard deviation distance along each axis. $\theta=0$ should correspond to the positive end of the major axis, and $\theta = \pi/2$ corresponds to the positive end of the minor axis of the ellipse, with radii equal to 2 times the square root of eigenvalue 1, and 2 times the square root of eigenvalue 2, respectively. Plot the ellipse using red dashed lines.

The plot should look like this:



Script ?

Save

Reset

MATLAB Documentation (<https://www.mathworks.com/help/>)

```

1 load fisheriris
2 D = meas; % [150 x 4] data feature matrix containing 4 features of 150 samples
3 % compute D_pca and eigvals using the pca function
4 [eigvecs,D_pca,eigvals] = pca(D); %, 'NumComponents',2);
5 %eigvecs
6 % Define D_pca_x and D_pca_y, the first two dimensions of the data in the D_pca feature space
7 D_pca_x = D_pca(:,1);
8 D_pca_y = D_pca(:,2);
9 % scatter plot the data with asterisks '*'
10 plot(D_pca_x, D_pca_y, '*');
11 % Define ellipse_x and ellipse_y, x and y coordinates for the PCA ellipsoid in the first 2 dimensions as a
12 theta = 2*pi*[0:100]/100;
13 ellipse_x = 2*sqrt(eigvals(1))*cos(theta);
14 ellipse_y = 2*sqrt(eigvals(2))*sin(theta);
15 %Plot the ellipse with red dashed lines 'r--'
16 hold on;
17 plot(ellipse_x, ellipse_y, 'r--')
18
19 %Use axis equal to correct aspect ratio
20 axis equal
21

```

Assessment: All Tests Passed

[Submit](#)

✓ Is D_pca correct?

✓ Is eigvals correct?

✓ Is D_pca_x correct?

✓ Is D_pca_y correct?

✓ Is ellipse_x correct?

✓ Is ellipse_y correct?

Output

